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$$f(x) = x^2$$

$$a_n = \langle f | \cos(nx) \rangle$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \cos(nx) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x^2 \cos(nx) dx$$

$$= \frac{2}{\pi} \left[\frac{x^2 \sin(nx)}{n} + \frac{2x \cos(nx)}{n^2} - \frac{2 \sin(nx)}{n^3} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \cdot \frac{2\pi(-1)^n}{n^2}$$

$$= \frac{4(-1)^n}{n^2}$$

$$b_n = \langle f | \sin(nx) \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \sin(nx) dx = 0$$

$$a_0 = \langle f | 1 \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 dx = \frac{2}{\pi} \int_0^{\pi} x^2 dx = \frac{2}{\pi} \left[\frac{x^3}{3} \right]_0^{\pi} = \frac{2\pi^2}{3}$$

$$\Rightarrow \hat{f}(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx)) = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos(nx)$$

References